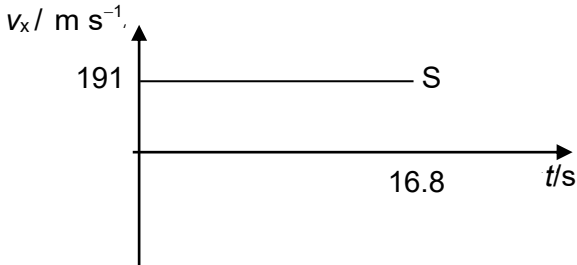
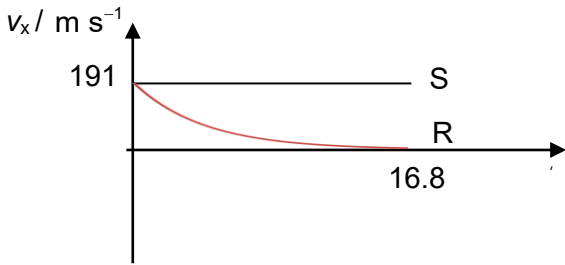


Qn	Suggested Solutions
2(a)(i)	$s_y = u_y t + \frac{1}{2} a_y t^2$ $x = 220 \sin(30^\circ)(16.8) + \frac{1}{2}(9.81)(16.8)^2$ $x = 3230 \text{ m}$
(ii)	$v_x = u_x = 220 \cos(30^\circ) = 190.5 \text{ m s}^{-1}$ $v_y = u_y + a_y t$ $v_y = 220 \sin(30^\circ) + 9.81(16.8) = 274.8 \text{ m s}^{-1}$ $v = \sqrt{v_x^2 + v_y^2}$ $v = \sqrt{190.5^2 + 274.8^2}$ $v = 334 \text{ m s}^{-1}$
	OR
	initial $E_k = \frac{1}{2} m (220)^2$
	loss in gravitational $E_p = mgh$ $= m(9.81)(3232)$
	final kinetic energy = loss in gravitational potential energy + initial kinetic energy $\frac{1}{2} m v^2 = m(9.81)(3232) + \frac{1}{2} m (220)^2$ $v = 334 \text{ m s}^{-1}$
(iii)	
(b)	

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